



# Ground Station Digital Clock Requirements & Work Plan

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*Ground Station Digital Clock – Requirements & Work Plan*  
*SpaceLab, Universidade Federal de Santa Catarina, Florianópolis – Brazil*

**Ground Station Digital Clock  
Requirements & Work Plan**  
*2026*

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## Nomenclature

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|        |                                              |
|--------|----------------------------------------------|
| BOM    | Bill of Materials                            |
| CI/CD  | Continuous Integration / Continuous Delivery |
| DRC    | Design Rules Check (KiCad)                   |
| DST    | Daylight Saving Time                         |
| EDA    | Electronic Design Automation                 |
| ERC    | Electrical Rules Check (KiCad)               |
| GPIO   | General Purpose Input/Output                 |
| LED    | Light-Emitting Diode                         |
| MCU    | Microcontroller Unit                         |
| NTP    | Network Time Protocol (RFC 5905)             |
| PCB    | Printed Circuit Board                        |
| RS-485 | TIA/EIA-485 Differential Serial Standard     |
| RTC    | Real-Time Clock                              |
| RVM    | Requirements Verification Matrix             |
| UTC    | Coordinated Universal Time                   |



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# CHAPTER 1

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## Introduction

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This document specifies the requirements for a custom-built digital clock to be installed in the SpaceLab UFSC ground station and control room. The clock is inspired by the Pro-Digital RDI-2M industrial wall clock – a 6-digit, Ethernet-synchronised display with ambient temperature alternation – and must be designed, fabricated, and verified entirely in-house by the laboratory team.

The document also defines the work plan describing how the four-person team is organised, the tasks to be completed in each phase, and the expected deliverables per task.

## 1.1 Scope

The scope of this document covers:

- A full set of requirements for all engineering disciplines: functional, physical, environmental, operational, interface, design, and mechanical.
- The GitHub repository structure and kick-off actions required before design work begins.
- A phased work plan for two hardware engineers and two firmware engineers working in parallel.
- Design constraints and notes that complement the formal requirements.

## 1.2 Document Identification

Table 1.1: Document identification.

| Field           | Value                                                      |
|-----------------|------------------------------------------------------------|
| Document Number | SLB-GS-CLK-REQ-v1.0                                        |
| Title           | Ground Station Digital Clock –<br>Requirements & Work Plan |
| Organisation    | SpaceLab, UFSC                                             |
| Location        | Florianópolis, Santa Catarina, Brazil                      |
| Version         | 1.0                                                        |
| Date            | 2026                                                       |
| Status          | Draft                                                      |

## 1.3 Reference Documents

Table 1.2: Reference documents.

| Ref.  | Identifier                | Title / Description                                                                                                                                     |
|-------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| [RD1] | Pro-Digital Orc.<br>52359 | Quotation No. 52359 dated 13 Nov 2025.<br>Commercial reference pricing for models<br>RDI-2P, RDI-2M, and RDI-2G.                                        |
| [RD2] | RDI Datasheet             | Pro-Digital Industrial Wall Clock<br>datasheet – operating modes, networking<br>options, dimensions, and model<br>comparison table.                     |
| [RD3] | SUP Datasheet             | Pro-Digital Suporte para<br>RDI/Cronômetros – bracket models<br>SUP-P-30, SUP-M-50, and SUP-G-60<br>with compatible clock models and<br>dimensions X/Y. |

## CHAPTER 2

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### Team Structure and Roles

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The project is staffed by four team members divided into two sub-teams. Table 2.1 summarises the team assignment and primary responsibilities.

Table 2.1: Team structure and responsibilities.

| Sub-team | Members                                  | Primary Responsibilities                                                                                                                                            |
|----------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hardware | Rian Carlos da Silva<br><b>Member X</b>  | Schematic design (KiCad), PCB layout, component selection, BOM drafting, enclosure fabrication, bracket installation, and power-on bring-up.                        |
| Firmware | Anthon Porath Gretter<br><b>Member Y</b> | Peripheral drivers, NTP client, RTC integration, display refresh, RS-485 communication, temperature display, brightness control, and build system / CI/CD pipeline. |



## CHAPTER 3

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### GitHub Repository Setup

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Before any design work begins, the team shall create a GitHub repository under the SpaceLab UFSC organisation. The following actions are **required** at kick-off:

- **Replicate the lab’s folder structure.** Browse existing SpaceLab repositories to identify the standard folder hierarchy used across projects.
- **Continue the base branch structure:** `main` (stable releases), `dev_firmware`, `dev_hardware`, `dev_mechanical`, `dev_doc`.
- **Add a `.gitignore`** appropriate for KiCad projects (exclude fabrication outputs and local caches), and a basic `README.md`.

### 3.1 Recommended Folder Structure

Table 3.1 lists the recommended top-level directories and files for the repository.

Table 3.1: Recommended repository folder structure.

| Path                             | Description                                                                                                                  |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <code>/.github/workflows/</code> | GitHub Actions CI: build check, KiCad ERC/DRC scripts.                                                                       |
| <code>/doc/</code>               | Architecture block diagram, firmware architecture document, datasheet references.                                            |
| <code>/hardware/</code>          | KiCad project root. Contains <code>.kicad_pro</code> , schematic sheets, PCB file.                                           |
| <code>/hardware/bom/</code>      | Bill of Materials (CSV and XLSX). Updated after each component selection iteration.                                          |
| <code>/hardware/outputs/</code>  | Fabrication outputs: Gerbers, drill files, assembly drawings.                                                                |
| <code>/firmware/</code>          | Firmware source tree. Follows the structure of the chosen SDK (e.g. ESP-IDF <code>components/</code> ).                      |
| <code>/firmware/tests/</code>    | Unit tests for peripheral drivers, NTP parser, and DST logic.                                                                |
| <code>/LICENSE</code>            | Creative Commons Attribution Share Alike 4.0 International for documentation and GNU General Public License v2.0 for coding. |
| <code>/README.md</code>          | Project overview, hardware version, build instructions, programming guide, team contacts.                                    |

## CHAPTER 4

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### Requirements

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All requirements follow the format used in other SpaceLab subsystem documents. Each requirement carries a unique identifier, a type, a priority level (High / Low), a plain-English description, the rationale for inclusion, and the verification method.

#### 4.1 Requirement Types and Verification Methods

Table 4.1: Requirement types.

| Type        | Description                                                              |
|-------------|--------------------------------------------------------------------------|
| Mission     | High-level objectives the clock must fulfil for the laboratory.          |
| Functional  | Observable behaviours the clock shall exhibit during operation.          |
| Physical    | Constraints on dimensions, mass, materials, and visual appearance.       |
| Environment | Operating conditions (temperature, humidity, power environment).         |
| Operational | Power, brightness, user-interaction, and status-indication requirements. |
| Interface   | External electrical/data interfaces the clock shall provide.             |
| Design      | Constraints on tools, architecture, and implementation choices.          |
| Mechanical  | Mounting, enclosure fastening, and structural integrity requirements.    |

Table 4.2: Verification methods.

| Method        | Description                                                                    |
|---------------|--------------------------------------------------------------------------------|
| Inspection    | Visual examination or review of design artefacts (schematics, PCB, documents). |
| Testing       | Measurement or functional test performed on the physical hardware.             |
| Analysis      | Engineering calculation, simulation, or analytical model.                      |
| Demonstration | Witnessed operation of a specific feature without instrumentation.             |

## 4.2 Requirements Table

The full requirements list is presented in Table 4.3 (landscape, next pages).



Table 4.3: Ground station clock requirements.

| ID             | Type       | Level | Description                                                                                                                   | Rationale                                                                             | Verification |
|----------------|------------|-------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------|
| GS-CLK-REQ-000 | Mission    | High  | The clock shall be designed and fabricated entirely in-house by the laboratory team.                                          | Ensures full technical ownership, traceability, and reproducibility within the lab.   | Inspection   |
| GS-CLK-REQ-001 | Mission    | High  | The clock shall serve as the primary time reference displayed in the ground station / control room.                           | Provides operators with a clear, unambiguous time reference during satellite passes.  | Inspection   |
| GS-CLK-REQ-100 | Functional | High  | The clock shall display hours, minutes, and seconds simultaneously using a 6-digit format (HH:MM:SS).                         | Seconds resolution is required for precise tracking and command scheduling.           | Testing      |
| GS-CLK-REQ-101 | Functional | High  | The clock shall synchronise its internal time via NTP over an Ethernet connection.                                            | Ensures traceable UTC synchronisation without manual adjustment.                      | Testing      |
| GS-CLK-REQ-102 | Functional | Low   | The clock shall alternately display ambient temperature alongside the current time.                                           | Provides environmental monitoring directly on the ground station display.             | Testing      |
| GS-CLK-REQ-103 | Functional | Low   | The clock shall apply automatic Daylight Saving Time (DST) adjustment according to the Brazilian official calendar.           | Avoids manual reconfiguration every DST transition.                                   | Testing      |
| GS-CLK-REQ-104 | Functional | Low   | The clock shall operate autonomously (free-running from its internal RTC) when the Ethernet network is unavailable.           | Guarantees continued display even during network outages.                             | Testing      |
| GS-CLK-REQ-105 | Functional | Low   | The clock shall retain the correct time for at least 72 hours during a power outage using a backup battery or supercapacitor. | Prevents time loss after brief power interruptions common in laboratory environments. | Testing      |

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Table 4.3 – continued from previous page

| ID             | Type        | Level | Description                                                                                                               | Rationale                                                                                                        | Verification |
|----------------|-------------|-------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|--------------|
| GS-CLK-REQ-200 | Physical    | High  | The clock enclosure shall not exceed 600 mm × 160 mm × 80 mm (W × H × D).                                                 | Sized to fit the available wall space in the ground station, inspired by the Pro-Digital RDI-2M reference clock. | Inspection   |
| GS-CLK-REQ-201 | Physical    | High  | The display shall use large 7-segment LED digits with a minimum digit height of 57 mm.                                    | Ensures visibility up to 40 m across the control room.                                                           | Inspection   |
| GS-CLK-REQ-202 | Physical    | Low   | The enclosure shall be fabricated from metal (aluminium or steel) with a black or dark-grey finish.                       | Provides electromagnetic shielding and a professional appearance matching lab equipment.                         | Inspection   |
| GS-CLK-REQ-203 | Physical    | Low   | The total mass of the assembled clock (including enclosure and bracket) shall not exceed 3 kg.                            | Keeps installation simple with standard wall anchors.                                                            | Inspection   |
| GS-CLK-REQ-300 | Environment | High  | The clock shall operate correctly in the temperature range 0 °C to 50 °C.                                                 | Covers the expected indoor operating envelope of the ground station room.                                        | Testing      |
| GS-CLK-REQ-301 | Environment | Low   | The clock shall include an onboard temperature sensor with a measurement range of 0 °C to 55 °C and accuracy $\leq 1$ °C. | Enables the alternating temperature display feature (REQ-102).                                                   | Testing      |
| GS-CLK-REQ-400 | Operational | High  | The clock shall operate on 100–240 V AC, 50/60 Hz (universal mains input).                                                | Allows use anywhere in the laboratory building without power adapters.                                           | Testing      |
| GS-CLK-REQ-401 | Operational | Low   | The clock shall consume no more than 12 W under normal operation.                                                         | Keeps the device within the power budget allocation for lab instruments.                                         | Analysis     |

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Table 4.3 – continued from previous page

| ID             | Type        | Level | Description                                                                                                                                       | Rationale                                                                                                                                    | Verification |
|----------------|-------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| GS-CLK-REQ-402 | Operational | Low   | The display brightness shall be adjustable (at minimum: high / medium / low) through a front-accessible or firmware-configurable setting.         | Reduces eye strain during night-shift operations and extends LED lifetime.                                                                   | Demo         |
| GS-CLK-REQ-403 | Operational | Low   | The clock shall provide a status indicator (LED or segment) that shows NTP synchronisation state.                                                 | Allows operators to verify time traceability at a glance.                                                                                    | Inspection   |
| GS-CLK-REQ-500 | Interface   | High  | The clock shall include a 10/100BASE-T Ethernet port for NTP time synchronisation and remote configuration.                                       | Standard interface available on all laboratory networking equipment.                                                                         | Testing      |
| GS-CLK-REQ-501 | Interface   | Low   | The clock shall include an RS-485 serial port to allow daisy-chaining of additional slave clocks in future.                                       | Enables low-cost network expansion without additional Ethernet infrastructure, consistent with the Pro-Digital reference architecture [RD2]. | Inspection   |
| GS-CLK-REQ-502 | Interface   | Low   | The clock shall expose a USB or JTAG/SWD debug port accessible without dismantling the enclosure.                                                 | Facilitates firmware updates and debugging by the firmware team.                                                                             | Inspection   |
| GS-CLK-REQ-600 | Design      | High  | All schematics and PCB layouts shall be designed in KiCad (version 8 or later).                                                                   | Aligns with the laboratory's existing EDA tool standardisation.                                                                              | Inspection   |
| GS-CLK-REQ-601 | Design      | High  | All design files (KiCad project, firmware, documentation) shall be version-controlled in a GitHub repository under the laboratory's organisation. | Ensures traceability, peer review, and long-term maintainability.                                                                            | Inspection   |

Continued on next page...

Table 4.3 – continued from previous page

| ID             | Type       | Level | Description                                                                                                                                                           | Rationale                                                                                                              | Verification |
|----------------|------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|--------------|
| GS-CLK-REQ-602 | Design     | High  | The microcontroller shall include sufficient GPIO to drive all display segments and interfaces without external bus expanders.                                        | Simplifies the design and reduces single-point-of-failure risks.                                                       | Analysis     |
| GS-CLK-REQ-603 | Design     | Low   | The PCB shall include a dedicated real-time clock (RTC) IC with a coin-cell or supercapacitor backup circuit.                                                         | Satisfies REQ-105 (72-hour time retention during power loss).                                                          | Testing      |
| GS-CLK-REQ-604 | Design     | Low   | The firmware shall be developed in C/C++ using a standard embedded framework (e.g. STM32CubeHAL, ESP-IDF, etc).                                                       | Leverages existing team expertise and the available library ecosystem.                                                 | Inspection   |
| GS-CLK-REQ-605 | Design     | Low   | The PCB shall include current and voltage sense points on the main power rail for debugging.                                                                          | Facilitates power-consumption measurement during verification.                                                         | Inspection   |
| GS-CLK-REQ-700 | Mechanical | High  | The clock shall be mounted using a Pro-Digital SUP-M-50 compatible ceiling/wall bracket, or a custom bracket with equivalent load capacity and adjustable tilt angle. | Replicates the reference installation from the Pro-Digital SUP datasheet [RD3] (X = 20 cm, Y = 42 cm for RDI-2M size). | Inspection   |
| GS-CLK-REQ-701 | Mechanical | High  | The mounting bracket shall support vertical and horizontal position adjustment of $\pm 30^\circ$ without tools.                                                       | Allows optimal viewing angle for all operators seated at the consoles.                                                 | Demo         |
| GS-CLK-REQ-702 | Mechanical | Low   | The enclosure shall use captive screws or quarter-turn fasteners for the front or rear access panel.                                                                  | Reduces maintenance time and risk of losing hardware during servicing.                                                 | Inspection   |
| GS-CLK-REQ-703 | Mechanical | Low   | All PCBs shall be secured inside the enclosure with stand-offs and locking hardware to withstand vibrations typical of laboratory environments.                       | Prevents PCB movement that could cause intermittent contacts or damage.                                                | Inspection   |

## CHAPTER 5

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### Work Plan

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The work plan is divided into four sequential phases. Hardware and firmware teams work in parallel during Phases 3 and 4. Phases 1 and 2 involve both teams together to ensure shared understanding of the system before splitting into parallel tracks.

Table 5.1 lists all tasks with their phase, assigned team, expected deliverable, and detailed instructions.

Table 5.1: Project work plan.

| Phase                             | Team | Task                              | Deliverable                                       | Notes / Instructions                                                                                                                                                                                           |
|-----------------------------------|------|-----------------------------------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phase 1 – Project Kick-off</b> | Both | Repository setup on GitHub        | Repository with folder structure, README, LICENSE | Inspect existing lab repositories to replicate folder structure.                                                                                                                                               |
|                                   | Both | Requirements review and baseline  | Signed-off requirements document                  | Each team member reads all requirements. Open GitHub Issues for any requirement deemed unclear or infeasible.                                                                                                  |
|                                   | Both | Development environment setup     | KiCad + toolchain installed on all machines       | HW team installs KiCad 8 and clones the repo. FW team installs the chosen SDK (e.g. ESP-IDF or STM32CubeIDE) and sets up a build pipeline (GitHub Actions).                                                    |
| <b>Phase 2 – Architecture</b>     | Both | System block diagram              | Block diagram committed to /docs/architecture/    | One session with all four members. Define major blocks: Power supply, MCU, RTC, Ethernet PHY, RS-485 transceiver, LED driver, 6-digit display, temperature sensor, backup battery.                             |
|                                   | HW   | Component selection and BOM draft | BOM draft in /hw/bom/                             | Select MCU, Ethernet PHY or built-in MAC, RS-485 transceiver, RTC IC, LED 7-segment modules, temperature sensor. Search in other lab projects for component recommendation. Verify availability and lead time. |
|                                   | FW   | Firmware architecture document    | FW architecture doc in /docs/firmware/            | Define RTOS tasks (or super-loop): NTP sync task, RTC management, display refresh ISR/task, RS-485 slave handler, temperature read, brightness control.                                                        |

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Table 5.1 – continued from previous page

| Phase                        | Team | Task                                           | Deliverable                                                          | Notes / Instructions                                                                                                                                                                                                                                                                           |
|------------------------------|------|------------------------------------------------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase 3 – Design             | HW   | Schematic design in KiCad                      | Schematic committed and passing ERC                                  | Divide schematic into sheets: Power, MCU core, Interfaces, Display & Drivers. Peer-review via pull request before PCB layout starts. Follow IPC-2221 clearance rules. Separate digital and power planes. Place RS-485 and Ethernet connectors at rear panel. Route display connector to front. |
|                              | HW   | PCB layout in KiCad                            | PCB layout passing DRC; Gerbers exported                             |                                                                                                                                                                                                                                                                                                |
|                              | FW   | Firmware skeleton and peripheral drivers       | Compiling FW skeleton with GPIO, UART, I <sup>2</sup> C, SPI drivers | Implement and unit-test each driver on an evaluation board before the custom PCB arrives. Commit tests to <code>/.../tests/</code> .                                                                                                                                                           |
|                              | FW   | NTP client and RTC integration                 | NTP sync demonstrated on eval board                                  | Implement NTPv4 over UDP. Validate UTC offset and DST handling for the <code>America/Sao_Paulo</code> timezone (UTC−3 / UTC−2 during DST).                                                                                                                                                     |
| Phase 4 – Integration & Test | HW   | PCB fabrication and assembly                   | Assembled and powered-on PCB                                         | Send Gerbers to fab. Assemble SMD components. Perform visual inspection before first power-on. Check all voltage rails with a multimeter before connecting the display.                                                                                                                        |
|                              | Both | Hardware–firmware integration                  | Clock displaying time on full hardware                               | Flash initial firmware. Verify display, NTP sync, temperature alternation, RS-485 loopback, and backup-battery retention. Log all defects as GitHub Issues.                                                                                                                                    |
|                              | Both | Enclosure fabrication and bracket installation | Clock installed on wall / bracket in ground station                  | Machine or order enclosure panels. Install PCB with stand-offs. Mount on SUP-M-50-compatible bracket. Verify tilt adjustment and cable management.                                                                                                                                             |

Continued on next page...

Table 5.1 – continued from previous page

| Phase | Team | Task                                              | Deliverable                                    | Notes / Instructions                                                                                                                         |
|-------|------|---------------------------------------------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
|       | Both | Requirements Verification Matrix (RVM) completion | Completed RVM committed to <code>/docs/</code> | Trace each requirement to its verification evidence (photos, test logs, measurements). Close all open GitHub Issues before final acceptance. |



## CHAPTER 6

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### Design Constraints and Notes

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The following constraints complement the formal requirements of Chapter 4 and provide additional implementation guidance for both teams.

1. **EDA tool.** All PCB design work shall be performed in KiCad version 8 or later. No proprietary EDA formats shall be committed to the repository.
2. **Timezone.** The firmware NTP client shall default to `America/Sao_Paulo` (UTC−3 / UTC−2 during DST). The DST transition dates shall be sourced from the official Brazilian government calendar.
3. **NTP protocol version.** The firmware shall implement NTPv4 as defined in RFC 5905, with at least two configurable server addresses (primary and fallback). Suggested defaults: `a.ntp.br` and `b.ntp.br` (NTP.br pool, Registro.br).
4. **Display visibility.** The chosen LED digit modules shall maintain adequate contrast in the ambient lighting conditions of the control room. A minimum contrast ratio of 3:1 between the illuminated segment and the enclosure background is recommended.
5. **RS-485 network.** When operating in slave mode on an RS-485 network, the firmware shall continue free-running timekeeping from the local RTC if the master signal is lost, consistent with the behaviour described in the Pro-Digital reference datasheet [RD2].